



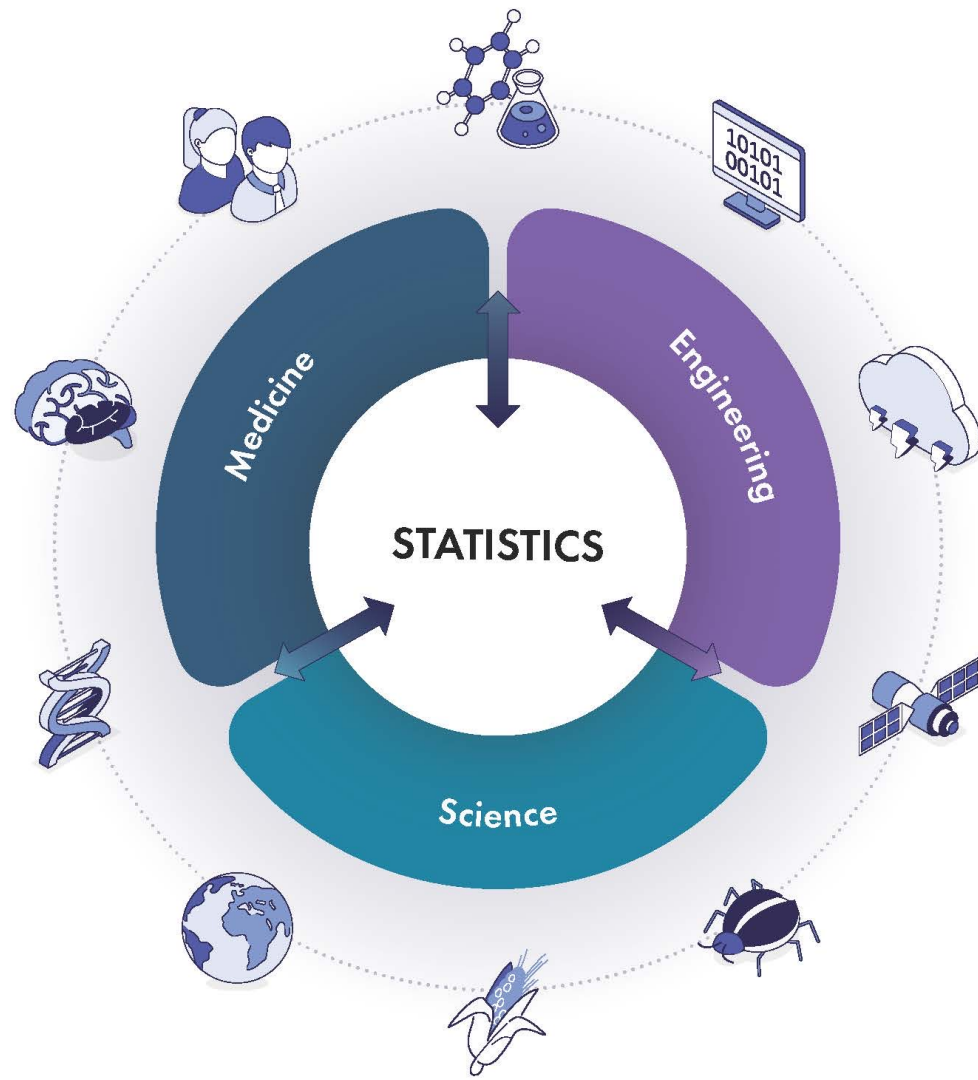
What does a statistician do?

Three research stories

TIAN ZHENG

PARTITION RETENTION, AUTOCLIMDS, GRADE THE GRADER

STAT UN 1001, FALL 2026



What is statistics for?

“Statistics is a wonderful discipline.”

“How do we translate information into knowledge?”

“How to turn data into decisions”

Stephen Senn, Dicing with Death

Three questions from my research

- 1** Why can a statistically significant variable predict poorly?
- 2** Can an AI agent assemble a trustworthy climate data workflow?
- 3** How do we know whether an AI grader judged an answer correctly?

Each problem asks how we generate evidence and whether our checks justify the conclusion.

Why significant variables aren't automatically good predictors

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Contributed by Herman Chernoff, September 17, 2015 (sent for review December 15, 2014)

Thus far, genome-wide association studies (GWAS) have been disappointing in the inability of investigators to use the results of identified, statistically significant variants in complex diseases to make predictions useful for personalized medicine. Why are significant variables not leading to good prediction of outcomes? We point out that this problem is prevalent in simple as well as complex data, in the sciences as well as the social sciences. We offer a brief explanation and some statistical insights on why higher significance cannot automatically imply stronger predictivity and illustrate through simulations and a real breast cancer example. We also demonstrate that highly predictive variables do not necessarily appear as highly significant, thus evading the researcher using

From the scientist's point of view there are two basic problems, complicated by the large size of the data set. These are variable selection and prediction. For variable selection, we wish to find a relatively small set of SNPs associated with the disease. For prediction we wish to find how a small set of such variables can be used to predict whether the subject has the disease. The size of the data set is such that the typical approach to variable selection has been to see how well correlated each SNP value is with the disease, and to keep only those for which the statistical significance was very high. Only recently has there been serious consideration of the possible interactions among two or more SNPs by some investigators. The prediction problem has typically been approached

Significance and prediction answer different questions

STATISTICAL SIGNIFICANCE

In the data at hand, how incompatible is the observed association with a specified chance model?

PREDICTION

For a new case, how close is the forecast to the outcome that actually occurs?

A small p-value can accompany a weak forecast. Prediction often depends on combinations, interactions, and validation on new data.

Partition retention searches for informative combinations

- 1 Start with a candidate set of variables and partition the cases by their joint values.
- 2 Score how much the outcome rates differ across the resulting groups.
- 3 Remove variables that do not improve the score. Repeat from many candidate sets.

Variables that repeatedly remain point to predictive combinations, including interactions that single-variable tests can miss.

BOTTLENECKS IN CLIMATE DATA SCIENCE

- ❑ Fragmented ecosystem: multiple portals, formats, agencies
- ❑ Retrieval-based workflows, not integration
- ❑ *“Complex and sometimes frustrating”* (Kempler et al, 2012)

Chapter 6

Data discovery, access and retrieval

S. Kempler (Auth./ed.)¹ with; G.G. Leptoukh¹, R.K. Kiang¹, R.P. Soebiyanto¹, D.Q. Tong², P. Ceccato³, S. Maxwell⁴, R.G. Rommel⁴, G.M. Jacques⁴, K.K. Benedict⁵, S.A. Morain⁵, P. Yang⁶, Q. Huang⁶, M.L. Golden⁷, R.S. Chen⁷, J.E. Pinzon⁸, B. Zaitchik⁹, D. Irwin¹⁰, S. Estes¹⁰, J. Luvall¹⁰, M. Wimberly¹¹, X. Xiao¹², K.M. Charland¹³, R.P. Stumpf¹⁴, Z. Deng¹⁵, C.E. Tilburg¹⁶, Y. Liu¹⁷, L. McClure¹⁸, & A. Huff¹⁹

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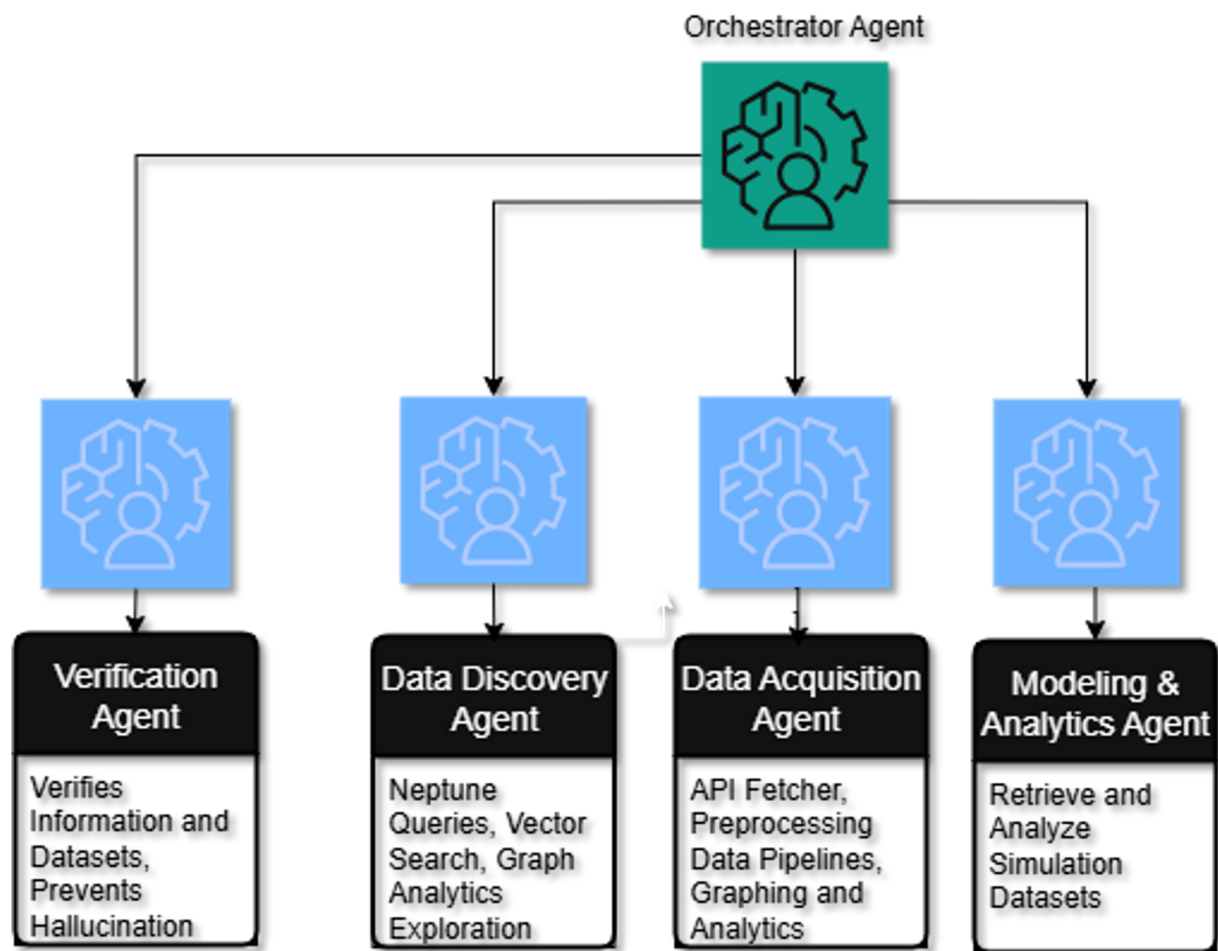
¹⁶University of New England, Biddeford, ME, US

¹⁷School of Public Health, Emory University, Atlanta, GA, US

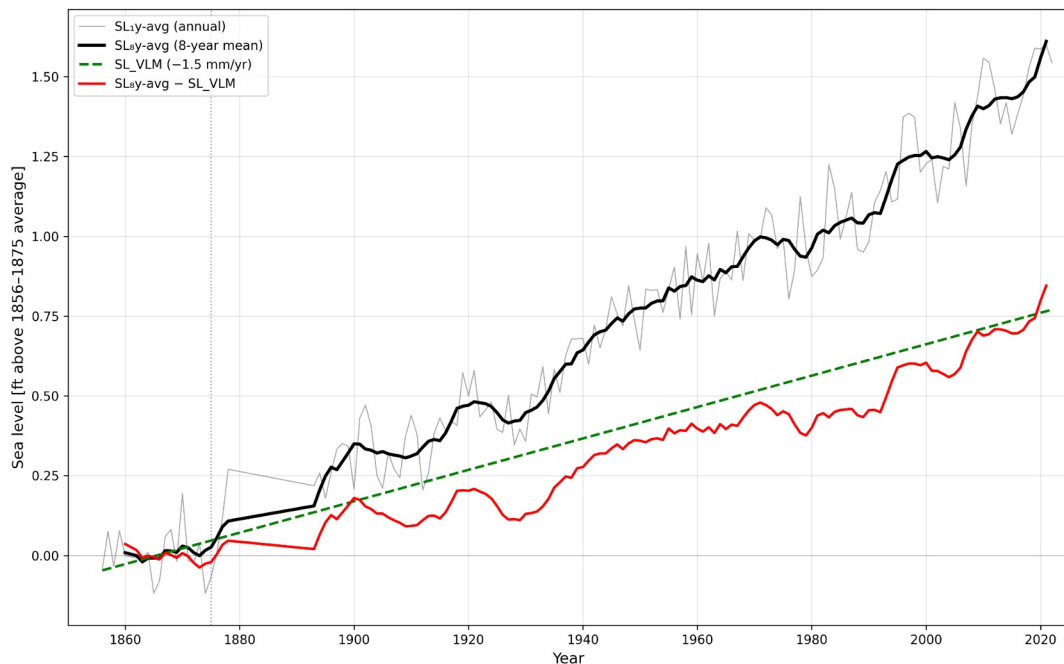
¹⁸University of Alabama, Birmingham, AL, US

¹⁹Battelle Memorial Institute, Columbus, OH, US

Kempler (Author/editor), S. (2012). Data discovery, access and retrieval. In Environmental Tracking for Public Health Surveillance. CRC Press.



Sea Level Decomposition: The Battery, New York (1856–2022)



AutoClimDS Replicated using natural language queries only

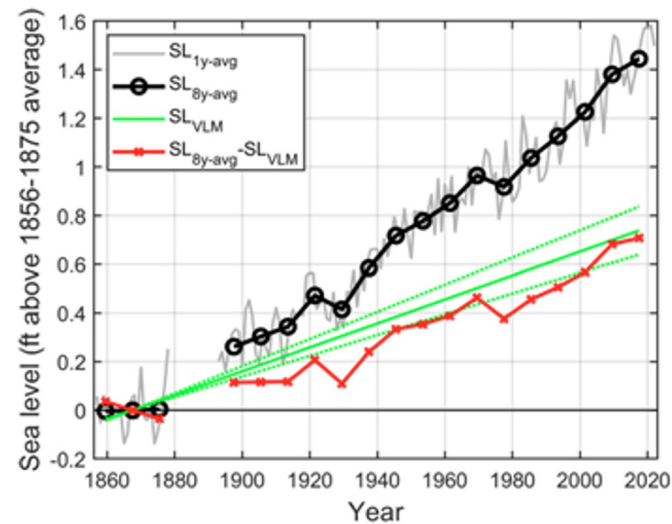


FIGURE 6 New York Harbor sea level from 1856 to 2022 with 1-year (gray) and 8-year (black) averaging, and as partitioned into change due to vertical land motion only (VLM; green) and rising sea levels only (red). A VLM estimate of -1.5 mm/yr ($\pm 0.2 \text{ mm}$ 95% confidence) for the Battery tide gauge is utilized.²⁵

Original Figure

What AutoClimDS tests

Can a natural-language question become a traceable climate analysis?

The workflow must find appropriate data and metadata.

It must retrieve and preprocess the data before running the analysis.

A verification agent checks sources and results throughout the process.

Scientists still decide whether the variables, assumptions, and comparison support the conclusion.

The Problem: *Grading* Is Its Own Failure Mode

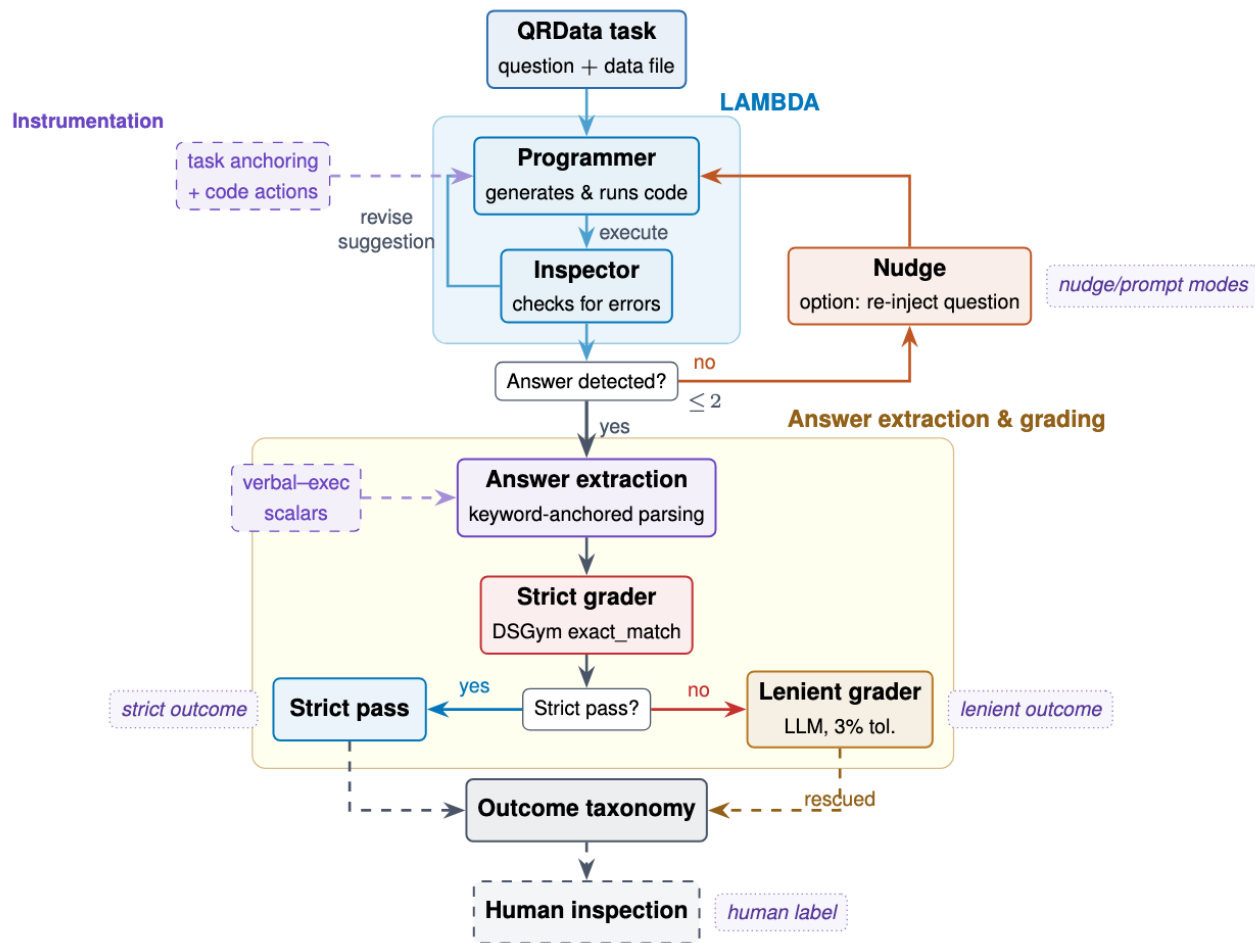
Case study: LAMBDA (multi-agent system) on
153 numerical QRData tasks from DSGym.

Agentic data-analysis systems

- produce rich, multi-step outputs: code, execution logs, intermediate statistics, and verbal diagnostics, etc.
- extracting the intended scalar answer from this noise is non-trivial, so grading itself becomes error-prone.

The Grader: DSGym's

- default `exact_match` scorer treats “0.2” and “0.20” as distinct, producing false negatives.
- Core tension: the richer the output, the harder it is to grade genuine agent–ground-truth disagreement must be separated from grading artifacts.



A grader must also be evaluated

36% to 97%

grading-run success after nudging

26% to 86%

strict-grader recall with keyword-anchored parsing

35% to 97%

lenient-grader recall after nudging

**Human labels help
separate agent errors
from grader errors.**

Benchmark: 153 numerical QRData tasks from DSGym

The common statistical question

What system produced the evidence, and what can the evidence support?

Partition retention

case and variable combinations shape forecasts for future outcomes

AutoClimDS

data sources and workflow produce a climate result

Grade the Grader

agent output and evaluation pipeline determine the reported score

These are the same questions we will ask all semester about cases, variables, samples, uncertainty, and claims.